

Prediction of Prices in Agriculture (PREPARE)

Date: 25 Feb 2026

Project Overview

The main idea behind our project is to solve farmer's price blindness. Right now, small and marginal farmers don't have accurate, timely price forecasts to understand market dynamics. We also want to provide reliable multi-day price prediction tools so that the government and regulators can plan better market interventions.

Semester 5 Progress

Last semester, we got the core parts of the project working. We developed a baseline model and did feature engineering. We also built a mobile app interface for the farmers and linked it to our model using FastAPI to get real-time crop price predictions for the next 15 days.

On the ML side, we started with a single-day GNN-RNN model and eventually moved to a spatiotemporal GAT-LSTM model to better capture market relationships and time trends. By the end of the semester, our main goals moving forward were to improve the multi-day model's accuracy and include a wider variety of seasonal and cash crops.

Please also find attached our presentation for last semester to get an idea about the app, model, and progress.

Semester 6 Progress

This semester, we've mostly been focusing on expanding our dataset and fixing up our pipeline. We wrote a scraping script using Python and Playwright to pull daily data from the Agmarknet portal for January 2023 to January 2026. We initially targeted onion, tomato, potato, sugarcane, and wheat to expand our crop variety.

However, after doing some basic preliminary EDA, we realized the sugarcane data was way too sparse to be useful, so we dropped it from our selection (we're sticking with wheat as our seasonal crop target). We also had two meetings with our mentor, Prof. Mayank Ratan Bhardwaj, where we brainstormed the best ways to impute the missing data, improve our scraping process, and plan out our next steps.

Next Steps

Over the next few weeks, our plan is to:

- Try out different imputation techniques on the dataset and see which methods are the most efficient.
- Use Earth Engine to add location and weather data to our scraped data.
- Group the mandis into clusters based on their production volume and test if this clustering affects the model's overall efficacy.